

Boss Challenge — Paper 22 (Class 7)

Transportation in Animals & Plants | Acids, Bases & Salts | Comparing Quantities | Perimeter & Area
One correct option each. Marking: +4 correct, -1 wrong, 0 blank. Write A/B/C/D on a separate sheet.

1. The red fluid that flows through the body of animals, carrying food, oxygen and wastes, is called:
(A) saliva
(B) urine
(C) bile
(D) blood
2. To convert the fraction $\frac{3}{4}$ into a percentage, we multiply it by:
(A) 4
(B) 10
(C) 1
(D) 100
3. Substances that taste sour and turn blue litmus paper red are called:
(A) bases
(B) acids
(C) metals
(D) salts
4. The perimeter of a rectangle is found using the formula:
(A) $4 \times \text{length}$
(B) $\text{length} \times \text{breadth}$
(C) $2 \times (\text{length} + \text{breadth})$
(D) $\text{length} + \text{breadth}$
5. The liquid part of blood, which is mostly water and carries the blood cells along, is called:
(A) haemoglobin
(B) plasma
(C) platelets
(D) lymph
6. 25% written as a fraction in its simplest form is:
(A) $\frac{25}{10}$
(B) $\frac{1}{2}$
(C) $\frac{1}{4}$
(D) $\frac{1}{5}$
7. Substances that taste bitter, feel soapy and turn red litmus paper blue are called:
(A) salts
(B) acids
(C) bases
(D) sugars

8. The area of a rectangle is found using the formula:

- (A) $2 \times (\text{length} + \text{breadth})$
- (B) $\text{length} \times \text{breadth}$
- (C) $\text{length} + \text{breadth}$
- (D) $4 \times \text{side}$

9. The blood cells that carry oxygen, coloured red by the pigment haemoglobin, are the:

- (A) platelets
- (B) nerve cells
- (C) red blood cells
- (D) white blood cells

10. Finding 10% of 250 gives:

- (A) 25
- (B) 2.5
- (C) 50
- (D) 250

11. The special paper or dye used to tell whether a substance is acidic or basic is called a/an:

- (A) filter
- (B) magnet
- (C) thermometer
- (D) indicator

12. A rectangular field of crops is 30 m long and 20 m wide. Its area is:

- (A) 100 square metres
- (B) 600 square metres
- (C) 600 metres
- (D) 50 square metres

13. The blood cells that protect the body by fighting against germs and disease are the:

- (A) white blood cells
- (B) platelets
- (C) plasma
- (D) red blood cells

14. A bottle of dilute acid contains 20% acid by volume in 500 mL of liquid. The volume of pure acid in it is:

- (A) 200 mL
- (B) 20 mL
- (C) 100 mL
- (D) 400 mL

15. Litmus, the common indicator, is obtained from a plant-like organism called a:
- (A) mango
 - (B) lichen
 - (C) rose
 - (D) mushroom
16. The perimeter of a square of side 7 cm is:
- (A) 49 cm
 - (B) 7 cm
 - (C) 14 cm
 - (D) 28 cm
17. The muscular organ that pumps blood all around the body without ever resting is the:
- (A) heart
 - (B) brain
 - (C) lungs
 - (D) stomach
18. The ratio 2 : 5 written as a percentage is:
- (A) 20%
 - (B) 25%
 - (C) 40%
 - (D) 50%
19. Turmeric, used as a natural indicator, turns which colour when added to a base?
- (A) black
 - (B) green
 - (C) blue
 - (D) red
20. A square tile measures 6 cm along each side. The surface area of one such tile is:
- (A) 36 square cm
 - (B) 24 square cm
 - (C) 36 cm
 - (D) 12 square cm
21. The thin-walled blood vessels where the actual exchange of food, oxygen and wastes with the body cells happens are the:
- (A) veins
 - (B) capillaries
 - (C) nerves
 - (D) arteries

22. If 60% of a class of 40 students passed a test, the number who passed is:

- (A) 24
- (B) 30
- (C) 16
- (D) 60

23. When an acid and a base are mixed in the right amounts, they cancel each other to form a salt and water. This reaction is called:

- (A) condensation
- (B) evaporation
- (C) respiration
- (D) neutralisation

24. The area of a triangle is found using the formula:

- (A) $\frac{1}{2} \times \text{base} \times \text{height}$
- (B) $2 \times \text{base} \times \text{height}$
- (C) $\text{base} + \text{height}$
- (D) $\text{base} \times \text{height}$

25. Blood vessels that carry blood AWAY from the heart, with thick, elastic walls, are called:

- (A) capillaries
- (B) arteries
- (C) nerves
- (D) veins

26. A shirt marked at 800 rupees is sold at a 25% discount. The discount in rupees is:

- (A) 775
- (B) 200
- (C) 25
- (D) 600

27. The sour taste of curd and lemon juice is due to the presence of:

- (A) acids
- (B) bases
- (C) salts
- (D) oils

28. A triangle has a base of 10 cm and a height of 6 cm. Its area is:

- (A) 16 square cm
- (B) 30 square cm
- (C) 60 square cm
- (D) 30 cm

- 29.** Blood vessels that carry blood BACK towards the heart, and which have valves to stop backflow, are called:
- (A) capillaries
 - (B) tendons
 - (C) arteries
 - (D) veins
- 30.** A resting heart beats 70 times a minute, and during exercise it beats 105 times a minute. The ratio of resting to exercise rate, in simplest form, is:
- (A) 1 : 2
 - (B) 7 : 10
 - (C) 2 : 3
 - (D) 3 : 2
- 31.** The acid present in the stomach that helps digest food is:
- (A) carbonic acid
 - (B) acetic acid
 - (C) hydrochloric acid
 - (D) citric acid
- 32.** A leaf is roughly rectangular, 8 cm long and 3 cm wide. The surface area through which it can transpire on one side is about:
- (A) 24 square cm
 - (B) 22 square cm
 - (C) 24 cm
 - (D) 11 square cm
- 33.** Each single throb of an artery, felt where it runs close to the skin, is called the:
- (A) reflex
 - (B) nerve
 - (C) pulse
 - (D) clot
- 34.** To find the Simple Interest on a sum of money, we multiply the principal, the rate per year and the:
- (A) temperature, then subtract 100
 - (B) time in years, then divide by 100
 - (C) rate again, then multiply by 100
 - (D) number of coins, then add 100
- 35.** A common base used in the laboratory, also found in window-cleaning liquids, is:
- (A) vinegar
 - (B) sodium hydroxide
 - (C) citric acid
 - (D) hydrochloric acid

- 36.** The distance all the way around the edge of any flat shape is called its:
- (A) perimeter
 - (B) volume
 - (C) diameter
 - (D) area
- 37.** The number of times a healthy resting person's heart beats in one minute is closest to:
- (A) about 72 times
 - (B) about 1000 times
 - (C) about 300 times
 - (D) about 5 times
- 38.** The Simple Interest on 2000 rupees for 1 year at 5% per year is:
- (A) 1000 rupees
 - (B) 100 rupees
 - (C) 5 rupees
 - (D) 2100 rupees
- 39.** When an acid reacts completely with a base, the products are always a salt and:
- (A) an acid
 - (B) a metal
 - (C) water
 - (D) oxygen
- 40.** The amount of surface that a flat shape covers is called its:
- (A) area
 - (B) height
 - (C) mass
 - (D) perimeter
- 41.** The waste liquid produced when the kidneys filter the blood, and later passed out of the body, is:
- (A) plasma
 - (B) urine
 - (C) sweat
 - (D) bile
- 42.** A solution is made by mixing 30 mL of acid with 70 mL of water. The percentage of acid in the mixture is:
- (A) 3%
 - (B) 70%
 - (C) 30%
 - (D) 100%

43. An ant sting causes burning because the ant injects an acid. Rubbing a mild base such as baking soda on it helps because the base:

- (A) makes the skin colder
- (B) neutralises the acid
- (C) adds more acid to the skin
- (D) washes germs away

44. A square plot of land has a side of 25 m. The length of fencing needed to go right around it is:

- (A) 25 m
- (B) 625 m
- (C) 100 m
- (D) 50 m

45. The bean-shaped organs that filter wastes from the blood to form urine are the:

- (A) heart
- (B) kidneys
- (C) lungs
- (D) liver

46. If the price of a notebook rises from 20 rupees to 25 rupees, the percentage increase is:

- (A) 125%
- (B) 20%
- (C) 25%
- (D) 5%

47. Bee stings are acidic, while wasp stings are basic. A wasp sting is best soothed by rubbing on a mild:

- (A) base such as baking soda
- (B) salt water only
- (C) more wasp venom
- (D) acid such as vinegar

48. One square metre is equal to how many square centimetres?

- (A) 1000
- (B) 100
- (C) 10000
- (D) 1000000

49. In tall plants, water and dissolved minerals are carried UP from the roots to the leaves through tubes called:

- (A) stomata
- (B) phloem
- (C) roots hairs
- (D) xylem

50. A fruit seller buys mangoes for 80 rupees and sells them for 100 rupees. The profit percent is:
- (A) 100%
 - (B) 80%
 - (C) 25%
 - (D) 20%
51. A substance that is neither acidic nor basic, like pure water, is described as:
- (A) sour
 - (B) strongly acidic
 - (C) neutral
 - (D) strongly basic
52. A rectangular sheet is 12 cm long and 5 cm wide. Its perimeter is:
- (A) 24 cm
 - (B) 34 cm
 - (C) 60 cm
 - (D) 17 cm
53. In a plant, the food made in the leaves is carried to all other parts through tubes called:
- (A) veins of the heart
 - (B) stomata
 - (C) xylem
 - (D) phloem
54. Half of all the marks in an exam is the same as a percentage of:
- (A) 25%
 - (B) 75%
 - (C) 100%
 - (D) 50%
55. Soil that is too acidic for crops can be treated by the farmer by adding:
- (A) lime (a base) to the soil
 - (B) more acid to the soil
 - (C) plenty of sugar
 - (D) common salt
56. If the side of a square is doubled, its perimeter becomes:
- (A) twice as large
 - (B) four times as large
 - (C) the same as before
 - (D) half as large
57. The loss of water as vapour from the leaves of a plant, mainly through the stomata, is called:
- (A) respiration
 - (B) condensation
 - (C) germination
 - (D) transpiration

58. Air is about 78% nitrogen. In a sealed jar holding 50 units of air, the units that are nitrogen are about:

- (A) 50
- (B) 78
- (C) 22
- (D) 39

59. Litmus paper is described as a natural indicator. When dipped in lemon juice, blue litmus will turn:

- (A) red
- (B) green
- (C) black
- (D) stay blue

60. If the side of a square is doubled, its area becomes:

- (A) four times as large
- (B) the same as before
- (C) eight times as large
- (D) twice as large

61. The pull created by transpiration from the leaves is important because it helps the plant to:

- (A) draw water up from the roots
- (B) fight off germs
- (C) make its own food faster
- (D) store extra oxygen

62. The fraction $\frac{1}{5}$ expressed as a percentage is:

- (A) 5%
- (B) 15%
- (C) 50%
- (D) 20%

63. When a base turns red litmus blue and an acid turns blue litmus red, but the substance changes NEITHER colour, the substance is:

- (A) a strong acid
- (B) a strong base
- (C) a mixture of both at once
- (D) neutral

64. The area of a rectangle is 48 square cm and its length is 8 cm. Its breadth is:

- (A) 384 cm
- (B) 40 cm
- (C) 6 cm
- (D) 8 cm

65. Water and minerals from the soil first enter a plant through the fine projections on its roots called:
- (A) flowers
 - (B) root hairs
 - (C) fruits
 - (D) stomata
66. A 40-litre tank is 75% full of water. The volume of water it holds is:
- (A) 10 litres
 - (B) 75 litres
 - (C) 30 litres
 - (D) 40 litres
67. The white substance left behind when an acid and a base react and the water is then dried off is the:
- (A) indicator
 - (B) base
 - (C) acid
 - (D) salt
68. A square field has an area of 81 square metres. The length of each side is:
- (A) 9 m
 - (B) 18 m
 - (C) 81 m
 - (D) 40.5 m
69. The human heart is divided into how many chambers?
- (A) two
 - (B) one
 - (C) four
 - (D) six
70. In a survey, 3 out of every 4 students said they like science. As a percentage, this is:
- (A) 43%
 - (B) 34%
 - (C) 60%
 - (D) 75%
71. An antacid tablet relieves stomach acidity because it contains a mild:
- (A) salt only
 - (B) base
 - (C) sugar
 - (D) acid

72. The perimeter of a rectangular sheet of paper used for an experiment is 26 cm and its length is 8 cm. Its breadth is:
- (A) 5 cm
 - (B) 13 cm
 - (C) 18 cm
 - (D) 3 cm
73. During hard exercise, the heartbeat and pulse rate of a person usually:
- (A) increase
 - (B) stay exactly the same
 - (C) stop completely
 - (D) decrease
74. If 1 metre = 100 centimetres, then 40 cm written as a percentage of 1 metre is:
- (A) 400%
 - (B) 60%
 - (C) 4%
 - (D) 40%
75. Vinegar, used in the kitchen, is a dilute form of:
- (A) sodium hydroxide
 - (B) hydrochloric acid
 - (C) acetic acid
 - (D) lime water
76. A wire of length 24 cm is bent into a square. The length of each side of the square is:
- (A) 12 cm
 - (B) 6 cm
 - (C) 8 cm
 - (D) 24 cm
77. The tiny fragments in the blood that help it to clot and seal a cut are the:
- (A) nerve cells
 - (B) red blood cells
 - (C) white blood cells
 - (D) platelets
78. Two quantities are in the ratio 3 : 7 and the larger one is 35. The smaller one is:
- (A) 21
 - (B) 10
 - (C) 15
 - (D) 5

79. Citrus fruits such as oranges and lemons are sour because they contain:

- (A) acetic acid
- (B) citric acid
- (C) a base
- (D) hydrochloric acid

80. Two rectangular plots have the same area, but the first is long and narrow while the second is nearly square. Their perimeters are:

- (A) not necessarily equal
- (B) always four times the area
- (C) always exactly equal
- (D) always zero

81. Single-celled animals such as Amoeba do not need a transport system because they:

- (A) have a tiny heart of their own
- (B) never need oxygen at all
- (C) exchange materials directly with their surroundings
- (D) are made of xylem and phloem

82. A pure sample of a salt is found to be 90% pure in a 200 g packet. The mass of pure salt is:

- (A) 200 g
- (B) 20 g
- (C) 180 g
- (D) 90 g

83. Lime water is a well-known example of a:

- (A) strong acid
- (B) salt solution
- (C) neutral liquid
- (D) base

84. A path of width 1 m runs around the inside edge of a square room of side 10 m. The OUTER side of the square room is:

- (A) 8 m
- (B) 10 m
- (C) 12 m
- (D) 1 m

85. The transport of water in a tall tree from the roots up to the topmost leaves moves mainly in which direction?

- (A) in a circle inside one leaf
- (B) upward
- (C) sideways out of the trunk
- (D) downward only

86. A cricketer scores 45 runs out of his team's total of 180. His contribution as a percentage of the team total is:
- (A) 25%
 - (B) 20%
 - (C) 75%
 - (D) 45%
87. If too much acid and too much base are exactly balanced in a neutralisation, the final mixture is:
- (A) explosive
 - (B) still strongly acidic
 - (C) still strongly basic
 - (D) neutral
88. A rectangular garden is 15 m long and 10 m wide. The cost of fencing it at 5 rupees per metre is:
- (A) 250 rupees
 - (B) 50 rupees
 - (C) 750 rupees
 - (D) 125 rupees
89. Sweating helps the human body mainly by:
- (A) making the bones stronger
 - (B) adding oxygen to the blood
 - (C) digesting the food you eat
 - (D) removing some waste and cooling the body
90. Out of a 50-question Boss Challenge paper, a student answers 44 correctly. Their score as a percentage is:
- (A) 94%
 - (B) 44%
 - (C) 80%
 - (D) 88%
91. When testing many household items, baking soda solution would turn red litmus:
- (A) yellow
 - (B) red, with no change
 - (C) colourless
 - (D) blue
92. The area covered by 4 square tiles, each of side 10 cm, laid together is:
- (A) 160 square cm
 - (B) 400 square cm
 - (C) 40 square cm
 - (D) 100 square cm

93. Compared with arteries, the walls of veins are generally:
- (A) thinner
 - (B) made of bone
 - (C) filled with air
 - (D) much thicker
94. The marked price of a toy is 500 rupees and it is sold at 360 rupees. The discount percent is:
- (A) 36%
 - (B) 28%
 - (C) 72%
 - (D) 140%
95. The main reason we should NOT taste laboratory acids and bases to identify them is that they:
- (A) are always neutral
 - (B) can be corrosive and harmful
 - (C) taste too sweet
 - (D) would turn into water
96. A rope long enough to go exactly once around a square garden of side 12 m would have a length of:
- (A) 48 m
 - (B) 12 m
 - (C) 24 m
 - (D) 144 m
97. Plants need to absorb water and minerals from the soil mainly in order to:
- (A) breathe out carbon dioxide
 - (B) pump blood around
 - (C) make food and stay healthy
 - (D) change soil into stone
98. A dilute solution is labelled 15% salt by mass in a 200 g sample. The mass of pure salt it contains is:
- (A) 170 g
 - (B) 3 g
 - (C) 15 g
 - (D) 30 g
99. When carbon dioxide gas is bubbled through clear lime water, the lime water turns:
- (A) deep blue
 - (B) bright red
 - (C) milky white
 - (D) colourless and clear

100. A rectangular vegetable bed is 9 m long and 4 m wide. The area available for planting is:

- (A) 36 square metres
- (B) 36 metres
- (C) 26 square metres
- (D) 13 square metres